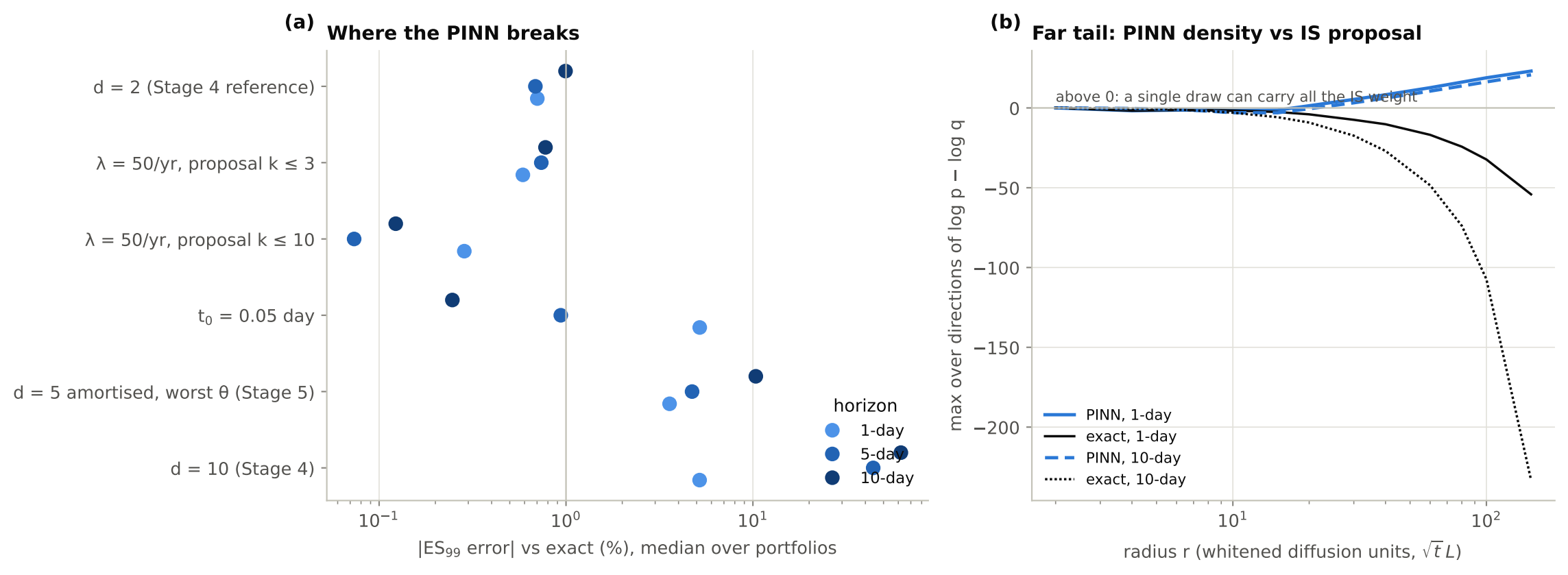
(a)
Where the PINN breaks
d = 2 (Stage 4 reference)
λ = 50/yr, proposal k ≤ 3
λ = 50/yr, proposal k ≤ 10
t0 = 0.05 day
d = 5 amortised, worst θ (Stage 5)
d = 10 (Stage 4)
horizon
1-day
5-day
10-day
10−1
100
101
|ES99 error| vs exact (%), median over portfolios
(b)
Far tail: PINN density vs IS proposal
above 0: a single draw can carry all the IS weight
max over directions of log p − log q
0
−50
−100
−150
−200
PINN, 1-day
exact, 1-day
PINN, 10-day
exact, 10-day
101
102
radius r (whitened diffusion units, √t L)